

一、填空题（每空 3 分，共 30 分）

1. $\frac{1}{3}$; 2. $P\{Z=0\}=0.25, P\{Z=1\}=0.75$; 3. e^{-1} ; 4. $f_X(x) = \begin{cases} \frac{\lambda}{(\lambda+x)^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}$;

5. $N(0,0,3,1,0)$; 6. $\theta+1$; 7. 0.99; 8. $F(1,1)$; 9. 0.64;

10. (0.0973, 0.3215). 【注：保留到小数点后三位即算对】

二解答题

11、（10 分） 解：设 $A = \{\text{甲最终获胜}\}$

$$D_1 = \{\text{第一轮掷骰子甲胜}\}, D_2 = \{\text{第一轮掷骰子乙胜}\}, D_3 = \{\text{第一轮掷骰子丙胜}\}$$

$$D_4 = \{\text{第一轮掷骰子没有分出胜负}\}. \text{ 则有}$$

$$P(D_1) = \frac{1}{6}, P(D_2) = \frac{5}{6} \times \frac{2}{6} = \frac{5}{18}, P(D_3) = \frac{5}{6} \times \frac{4}{6} \times \frac{3}{6} = \frac{5}{18}, P(D_4) = \frac{5}{6} \times \frac{4}{6} \times \frac{3}{6} = \frac{5}{18}$$

$$P(A|D_1) = 1, P(A|D_i) = 0 (i = 2, 3), P(A|D_4) = P(A) \quad \dots\dots 6 \text{ 分}$$

由全概率公式

$$P(A) = \sum_{i=1}^4 P(A|D_i)P(D_i) = P(D_1) + P(A)P(D_4) = \frac{1}{6} + \frac{5}{18}P(A)$$

$$\text{解得 } P(A) = \frac{3}{13}. \quad \dots\dots 10 \text{ 分}$$

12、（10 分） 解： 因 $X \sim P(\lambda)$ ，由题意 $P(X=1) = P(X=2)$ ，即有

$$\frac{\lambda}{1!} e^{-\lambda} = \frac{\lambda^2}{2!} e^{-\lambda} \quad \dots\dots\dots 3 \text{ 分}$$

故求得 $\lambda = 2$ ，即 $X \sim P(2)$. 于是每页无错误的概率为

$$p = P\{X=0\} = \frac{\lambda^0}{0!} e^{-\lambda} = e^{-2} \quad \dots\dots\dots 6 \text{ 分}$$

从而 $Y \sim b(10, e^{-2})$ ，故 Y 的分布律为

$$P\{Y=k\} = C_{10}^k e^{-2k} (1 - e^{-2})^{10-k}, k = 0, 1, \dots, 10 \quad \dots\dots\dots 10 \text{ 分}$$

13、（10 分） 解：由卷积公式有

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y)f_Y(y)dy = \int_0^{\infty} \lambda^2 y e^{-\lambda y} f_X(z-y)dy \quad \dots\dots 3 \text{ 分}$$

当 $z > 0$ 时

$$f_Z(z) = \lambda^3 \int_0^z y e^{-\lambda y} e^{-\lambda(z-y)} dy = \lambda^3 e^{-\lambda z} \int_0^z y dy = \frac{\lambda^3}{2} z^2 e^{-\lambda z}$$

所以

$$f_Z(z) = \begin{cases} \frac{\lambda^3}{2} z^2 e^{-\lambda z}, & z > 0 \\ 0, & z \leq 0 \end{cases} \quad \dots\dots 10 \text{ 分}$$

14、(10 分) 解:

$$EX = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy = 12 \int_0^1 x dx \int_0^x y^2 dy = 4 \int_0^1 x^4 dx = \frac{4}{5}$$

$$EY = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy = 12 \int_0^1 dx \int_0^x y^3 dy = 3 \int_0^1 x^4 dx = \frac{3}{5} \quad \dots\dots 4 \text{ 分}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy = 12 \int_0^1 x dx \int_0^x y^3 dy = 3 \int_0^1 x^5 dx = \frac{3}{6} = \frac{1}{2}$$

$$\text{cov}(X, Y) = E(XY) - (EX)(EY) = \frac{1}{2} - \frac{4}{5} \times \frac{3}{5} = \frac{1}{50} \quad \dots\dots 6 \text{ 分}$$

$$EY^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 f(x, y) dx dy = 12 \int_0^1 dx \int_0^x y^4 dy = \frac{12}{5} \int_0^1 x^5 dx = \frac{12}{5} \times \frac{1}{6} = \frac{2}{5}$$

$$D(X) = EX^2 - (EX)^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75}, \quad D(Y) = EY^2 - (EY)^2 = \frac{2}{5} - \left(\frac{3}{5}\right)^2 = \frac{1}{25}$$

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = \frac{\frac{1}{50}}{\sqrt{\frac{2}{75}}\sqrt{\frac{1}{25}}} = \frac{\sqrt{6}}{4} \quad \dots\dots 10 \text{ 分}$$

15、(15 分) 解: (1)

$$EX = \int_{-\infty}^{\infty} x f(x; \theta) = \int_1^{\infty} x \frac{\theta}{x^{\theta+1}} dx = \theta \int_1^{\infty} \frac{1}{x^{\theta}} dx = \frac{\theta}{\theta-1}$$

由矩估计法, 令 $\bar{X} = \frac{\theta}{\theta-1}$

解得 $\hat{\theta}_M = \frac{\bar{X}}{\bar{X}-1}$. \dots\dots 7 \text{ 分}

(2) 似然函数

$$L(\theta) = \prod_{i=1}^n \frac{\theta}{X_i^{\theta+1}} = \frac{\theta^n}{[\prod_{i=1}^n X_i]^{\theta+1}}, \quad \ln L(\theta) = n \ln \theta - (\theta+1) \sum_{i=1}^n \ln X_i$$

$$\frac{d \ln L(\theta)}{d \theta} = \frac{n}{\theta} - \sum_{i=1}^n \ln X_i \triangleq 0$$

$$\text{解得 } \hat{\theta}_L = \frac{n}{\sum_{i=1}^n \ln X_i}, \quad \dots\dots 15 \text{ 分}$$

16、(15 分) 解: 设 X 为失眠患者服甲种安眠药后的睡眠延长时数, Y 为失眠患者服乙种安眠药后的睡眠延长时数, 则有

$$X \sim N(\mu_1, \sigma^2), Y \sim N(\mu_2, \sigma^2)$$

我们要做的是对假设

$$H_0: \mu_1 = \mu_2 \quad \text{vs} \quad H_1: \mu_1 \neq \mu_2 \quad \dots\dots\dots 3 \text{ 分}$$

作显著性检验. 在显著性水平 α 下, H_0 的拒绝域为

$$|T| = \frac{|\bar{X} - \bar{Y}|}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \geq t_{1-\frac{\alpha}{2}}(n_1 + n_2 - 2) \quad \dots\dots\dots 8 \text{ 分}$$

这里 $n_1 = n_2 = 10$, 经计算 $\bar{X} = 2.33, S_1^2 = 4, \bar{Y} = 0.75, S_2^2 = 3.2$

$$S_w = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} = \sqrt{\frac{4+3.2}{2}} = 1.8974$$

$$|T| = \frac{|\bar{X} - \bar{Y}|}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{2.33-0.75}{1.8974 \sqrt{\frac{1}{10} + \frac{1}{10}}} = 1.862$$

因 $|T| = 1.862 < 2.1009 = t_{0.975}(18)$, 故接受 H_0 , 认为两种安眠药的疗效无显著差异. $\dots\dots\dots 15 \text{ 分}$